

12-Week Advanced Backend & System Design Roadmap

WEEK 1 — FOUNDATION SETUP

BUILD (60 min daily)

- Setup Node.js + TypeScript backend
- Create basic Express server
- Setup PostgreSQL locally
- Create basic API structure

SYSTEM DESIGN / CLOUD (30 min)

- What is backend architecture
- Client vs server
- HTTP basics
- REST API concept

DSA (30 min)

- **Pattern: TWO POINTERS (START)**
- Learn logic
- Solve: Two Sum (sorted), Palindrome check

WEEK 2 — DOCKER + API STRUCTURE

BUILD

- Dockerize backend
- Dockerize PostgreSQL
- Setup docker-compose
- Environment variables

SYSTEM DESIGN / CLOUD

- What is Docker
- Image vs container
- Why production uses containers

DSA

- **TWO POINTERS + HASHMAP INTRO**
- Remove duplicates
- Valid palindrome
- Two Sum (unsorted)

WEEK 3 — AWS DEPLOYMENT START

BUILD

- Deploy backend on EC2
- Run Docker on server
- Setup Nginx reverse proxy

SYSTEM DESIGN / CLOUD

- AWS EC2 basics
- SSH login
- Security groups

DSA

- **HASHMAP PATTERN**
- Group Anagrams
- Subarray Sum = K (basic)

WEEK 4 — DATABASE PRODUCTION

BUILD

- Move DB → AWS RDS
- Connect backend to RDS
- Fix connection pooling

SYSTEM DESIGN / CLOUD

- RDS vs local DB
- Connection pooling
- Latency concept

DSA

- **SLIDING WINDOW START**
- Longest substring without repeat
- Max sum subarray (fixed window)

WEEK 5 — REDIS (CACHING SYSTEM)

BUILD

- Add Redis caching
- Cache API responses
- Cache user sessions

SYSTEM DESIGN / CLOUD

- What is caching
- Cache hit vs miss
- TTL concept

DSA

- **SLIDING WINDOW + STACK START**
- Minimum window substring
- Valid parentheses

WEEK 6 — QUEUE SYSTEM

BUILD

- Add queue (BullMQ / RabbitMQ)
- Background email system
- Async jobs

SYSTEM DESIGN / CLOUD

- Queue concept
- Producer vs consumer
- Async processing

DSA

- **STACK + BINARY SEARCH**
- Next greater element
- Binary search basics

WEEK 7 — SYSTEM DESIGN WEEK

BUILD

- URL Shortener project module
- Basic chat system module

SYSTEM DESIGN / CLOUD

- Load balancing
- Rate limiting
- Stateless system idea

DSA

- **TREES INTRO**
- DFS traversal
- BFS level order

WEEK 8 — DATABASE OPTIMIZATION

BUILD

- Add indexing
- Optimize slow queries
- Fix DB performance issues

SYSTEM DESIGN / CLOUD

- Indexing concept
- Query execution plan
- Why queries become slow

DSA

- **RECURSION START**
- Subsets
- Permutations (basic)

WEEK 9 — AI INTEGRATION START

BUILD

- AI chat API
- LLM integration
- Basic assistant feature

SYSTEM DESIGN / CLOUD

- What is LLM
- Prompt engineering basics
- API cost concept

DSA

- **TREES + GRAPHS BASIC**
- Number of islands
- Tree height problem

WEEK 10 — ADVANCED AI FEATURE

BUILD

- Choose ONE: PDF chatbot OR Resume analyzer OR Smart search system

SYSTEM DESIGN / CLOUD

- Data flow in AI apps
- Chunking concept
- Embedding idea (basic)

DSA

- **GREEDY START**
- Activity selection
- Jump game

WEEK 11 — SYSTEM HARDENING

BUILD

- Rate limiting middleware
- Error handling system
- Logging system

SYSTEM DESIGN / CLOUD

- Failure handling
- Retry systems
- System resilience

DSA

- **BASIC DP START**
- Fibonacci
- Climbing stairs

WEEK 12 — FINAL PROJECT INTEGRATION

BUILD

- Final system: Auth, Redis, Queue, AI, AWS deployment, CI/CD, Logging

SYSTEM DESIGN / CLOUD

- Full architecture revision
- Scaling system overview

DSA

- **REVISION ONLY**
- Mix of all patterns